Yulin (Jason) Liu

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Technical Skills

Mechanical Design & Prototyping: SolidWorks, AutoCAD, Fusion 360, 3D Printing, CNC machining and fixture design (dovetail joints), MasterCAM, WARDJet waterjet
Robotics, Automation & Control Systems: LabVIEW, SMAC Actuator, Load Cell Calibration, Closed-Loop Control, RobotStudio, Siemens, Allen-Bradley, ROS, OpenCV, Linux, Arduino, C++, Python, SQL
Simulation & Analysis: Simulink, MATLAB, ANSYS, FEA
Additional Background: English, Mandarin, Cantonese; Clarinet (First-Chair, Kansas All-State Band); Rock Band Vocalist and Founder; Soccer (Varsity Captain)

Education

University of California San Diego — Sept. 2021 – June 2025
B.S., Mechanical Engineering (Control & Robotics) — *San Diego, CA*

Experience

Trane Technologies — July 2024 – Dec. 2024
Advanced Manufacturing Engineering Automation Co-op — *Pueblo, CO*

- **Automated Chiller Line**: Led design and commissioning of a turnkey robotic assembly line for shell-and-tube heat exchanger subassemblies, coordinating RFQs and vendor integration to meet production targets.
- **Standardized Work**: Optimized station layouts and workflows with operators, reducing cycle time by 20% under lean manufacturing guidelines.
- **Robotic Integration**: Configured FANUC M-710iC 12L robots with Allen-Bradley PLC/HMI, achieving 200 mm/s motion profiles and ±0.05 mm repeatability.

Projects

Biomechanical Culture System Cellxercise | *SMAC Integrator* — Jan. 2025 – Present

- **Bioreactor Design**: Developed a system delivering 20 N at 10% cyclic strain and 1 Hz to replicate physiologically relevant tensile conditions for tissue growth.
- **Control System**: Implemented closed-loop LabVIEW control with SMAC actuator and ATO load cell, yielding 0.01 mm positioning resolution and 5 µm measurement accuracy.
- **Modular Fabrication**: Fabricated sterilizable steel clamps and aluminum lift plates using MasterCAM, HAAS CNC mills, and WARDJet waterjet.

Autonomous Vehicles TranquiBot | *Project Lead* — Jan. 2024 – Present

- **Embedded Systems Architecture**: Architected an autonomous vehicle platform on Jetson Nano, integrating OAK-D depth camera, LD06 LiDAR, and Point One Fusion Engine GPS within ROS2 and the DonkeyCar framework.
- **Machine Learning & Perception**: Implemented OpenCV-based perception and TensorFlow inference for lane detection and waypoint generation, enabling real-time autonomous navigation from multimodal sensor inputs.
- **LLM-Driven Autonomy & Control**: Orchestrated ROS2 and Python middleware leveraging the ChatGPT API for high-level decision-making, translating perception outputs into motion commands executed via a VESC motor controller.

Robotic Lift Mechanism | *Linkage Designer* — Sept. 2022 – Dec. 2022

- **Design & Prototyping**: Generated SolidWorks models and laser-cut prototypes to validate four-bar lift kinematics.
- **Torque Optimization**: Refined a 1:4 double-gear system to boost lift capacity by 30%.
- **GD&T Analysis**: Performed GD&T evaluations and drafted detailed force, torque, and kinematic reports.

Research

Boomerang Research Laboratory | *Researcher* — Feb. 2023 – Present

- **Paper & Simulation**: Contributed to an AIAA paper on wing-tip deflection and built MATLAB airflow models with UWB tracking algorithms.
- **Experimental Validation**: Conducted 15+ field trials at six joint angles in low-wind conditions, comparing experimental results with simulations to assess the impact of joint angles.

Xtreme Materials Laboratory | *Researcher* — July 2022 – Jan. 2023

- **Composite Synthesis**: Synthesized high-purity composites via solvothermal methods and ±10 µm precision machining, increasing yield by 15%.
- **Purification Protocols**: Executed acid-washing, ultrasonication, and centrifugation, achieving 93% purity in a controlled nitrogen glove box environment.

Leadership

ASME UCSD Chapter | *Co-Chair* — Sept. 2023 – Present

- **Recruitment & Onboarding**: Oversaw hiring and integration of a 50-member chapter, instituting feedback loops to refine member experience.
- **Networking Initiatives**: Hosted Intern Mixer, Mentor–Mentee sessions, and MATLAB workshops, fostering communication and collaboration among engineering students.